

ABOUT

ENGINEER WITH A FOUNDATION IN AUTOMATION, MACHINE SOFTWARE, AND INDUSTRIAL SYSTEMS. I HAVE BUILT MY CAREER ACROSS MANUFACTURING, ROBOTICS, AND EMBEDDED TECHNOLOGIES. I VALUE CLARITY, AND CONTINUOUS IMPROVEMENT. I AM SEEKING A ROLE THAT OFFERS STRONGER OWNERSHIP AND THE OPPORTUNITY TO CONTRIBUTE TO IMPACTFUL ENGINEERING WORK.

EDUCATION

JAN 2016-17	MS, ELECTRICAL & COMPUTER ENGINEERING, (3.41/4)  NYU TANDON SCHOOL OF ENGINEERING
JUN 2010-14	BE, INSTRUMENTATION & CONTROLS, (7.04/10) SVIT, GUJARAT TECHNOLOGICAL UNIVERSITY

SKILLS

COMPUTER PROGRAMMING:	JAVA(PRIMARY), C/C++/CS (SECONDARY), GIT, PYTHON AUTOMATION, WEB(HTML, CSS, JS), DESKTOP(JAVAFX, AWT)
AUTOMATION:	MICRO-CONTROLLERS & SINGLE BOARD COMPUTERS,
INDUSTRIES SERVED:	MACHINERIES, MANUFACTURING, CONTROL PANELS, SOLAR ENERGY, EDUCATION

WORK EXPERIENCE

 SOFTWARE ENGINEER III - AUTOMATION, UNIVERSAL INSTRUMENTS, NY Sep 22 - present

- **RG Traceability Plus (2026)** : Enhanced archive naming, barcode integration, AVP traceability, and release-branch support for production customers.
- **UHA Customer Issue Resolution (2026)** : Resolved zeroing issues, PEC barcode failures, and PBI data defects; performed production runtime debugging.
- **SBS – Smart Barcode Scanning (2025–2026)** : Integrated scanner hardware; added ROI control, Live View app, traceability, error recovery, multi-bank support; completed 10,000+ cycle reliability testing.
- **Wafer AMS Project (2026)** : Implemented wafer find, alignment, placement, and multi-location workflows; delivered documentation, PRs, and machine testing.
- **Park Position Configuration (FUZ-6303)** : Added configurable park-position support; built machine-config UI; validated and released.
- **Wafer Substrate Mapping (FUZ-6293)** : Developed wafer mapping import + UI workflows; delivered customer demos; integrated mapping into production flow.
- **Xilinx Porting Project (FUZ-6274)** : Ported software from 4.4.0/4.4.1/4.5.0; added firmware support; completed platform validation.
- **HSWF Plus Features (FUZ-5805, 5810, 5811, 5812)** : Implemented heated spindle, heated bond process, gel-pack feeder support, and flipper enhancements for HSWF workflows.
- **Triangle / Specialized Fiducials (2025)** : Added isosceles triangle fiducial support; conducted machine testing; validated production readiness.
- **Buffer Tray Management (FUZ-4966, 5114, 5123)** : Implemented buffer tray support, corner fiducial find, verification, retry/continue logic, and tray-change detection.
- **Error Recovery Enhancements (FUZ-4948)** : Added new recovery events; extended MCOS error handling; improved operator recovery workflows for machine uptime.
- **C12 Import/Export (FUZ-4906)** : Implemented C12 import/export; synchronized categories, priorities, and buffer trays; completed full integration testing.
- **Feeder Database Teach Improvements (FUZ-4804)** : Added feeder selection UI; updated teach workflow; implemented SMP tray processing; improved backend logic for operator reliability.
- **Intel POI / Med-3 Program (2024)** : Delivered DFF handling, selective placement, alarms, buffer trays, patch/interposer categories, and priority avoidance; supported Release Rev13.
- **Traceability Barcode Enhancements (FUZ-4494, FUZ-4601)** : Added tray/lot traceability, barcode support, DTF feeder handling, and enhanced event logging + traceability outputs.
- **Nested / Interposer Tray Support (FUZ-4555)** : Implemented nested tray architecture; extended ProductConverter; added licensing; updated XML import/export for production release.
- **Multi-Rot Feeder Support (FUZ-4579)** : Added licensing + Win7 support; validated feeder operation; released full production functionality.
- **Musashi Error Monitoring (FUZ-3691)** : Built serial communication stack for Musashi dispensers; added product programming, parameter download, error-bit monitoring, and recovery logic.
- **Light Intensity Controller (LIC)** : Developed multi-channel LIC application; controller discovery; network connectivity; integrated illumination control into NightlyBuild.
- **AVP Fiducial System (FUZ-4214, FUZ-4415)** : Implemented new AVP fiducial type; added DB, XML/BMP gen-

eration, teach/find workflows, and correction processing for NPI + Production.

- **AVP Platform Development** : Ported AVP into Precision Bonder; built component architecture, ECS integration, DB support, VPP management, and image-collection workflows.

- **Manual Alignment Enhancement** : Corrected theta math and alignment workflow; fixed correction indexing; validated on production machines to eliminate alignment drift.

- **Dispense Inspection System** : Developed dispense-inspection VPPs; added reject/recovery logic; integrated inspection events with MCOS; improved dispense quality verification.

- **QBIT / Scan3D Integration (FUZ-3966)** : Integrated QBIT inspection into MCOS; built TCP interface; added reject/retry workflows; removed VPP delays; stabilized inspection triggering across machines.

- **Line Find Porting & Enhancement (FUZ-3763)** : Ported Padsite Line-Find into Precision Bonder; rewrote PEC math; fixed fiducial/corner/theta logic; added caliper controls; delivered full production validation.

- Develop and maintain software for the **Fuzion** platform.

- Implement **side-barcode scanning** using the Keyence SR-X300 for improved part traceability.

- Upgraded **AMS** (Accuracy Management System) from one-point to 13-location for higher accuracy on Wafer.

- Build **Selective Placement** with part buffering, enabling smarter pick/hold/place decisions.

- Cognex **Quick-build** integration in Fuzion platform

- Implemented **AVP** (Advanced Vision Processing) to allow user-defined fiducial finds.

- Tech-stack : · CS · C · CPP · Git · VxWorks · Sourcetree

🔧 AUTOMATION ENGINEER , NOR-CAL CONTROLS, CA

Jan 22 - Jul 22

- Developed automation software for utility-scale solar power plants

- Tech stack: Ignition SCADA · GE PLC · SEL-RTAC · Red Lion

🔧 SOFTWARE ENGINEER - MACHINE AUTOMATION, MINI-CIRCUITS, NY

Feb 18 - Nov 21

- Developed automation software for pick-and-place rotary machines with computer vision inspection.

- Implemented the new shopping cart system for the online store <https://www.minicircuits.com/>

- Developed enterprise application software for internal engineering teams.

- Tech stack: Java · JavaFX · HTML · CSS · JavaScript

🔧 DEVELOPER INTERN, REVMAX FLEET OPTIMIZATION, NY

Sep - Dec 17

- Developed firmware for the SidewalkIQ sensor using Doppler radar to monitor sidewalk activity.

- Deployed sensors at multiple NYC locations and validated field performance.

- Conducted R&D to improve data consistency and optimize storage and transfer to backend servers.

- Tech stack : · Arduino-C ·

🔧 INSTRUCTOR (ITEST PROGRAM), ROBOTICS & MECHATRONICS DPT., 🏫 NYU

May - Aug 17

- Mentored to Robotics Club Programming Team for Robotics competitions (First Robotics & Vex Robotics).

- Taught concepts of Computer Programming to Robotics Club Team.

🔧 LEAD INSTRUCTOR (CREST PROGRAM), CENTER FOR K12 STEM EDUCATION, 🏫 NYU

Sep 17 - Dec 17

- Training High School Interns with Computer Programming (Arduino C), IoT, Robotics & 3D-Printing activities.

- Led STEM projects including :

- Autonomous Car (Mouse in a Maze) - STM32 Board: https://github.com/sap586/RTES_Project

- Wireless Robot Controller - Arduino: <https://github.com/sap586/WirelessRobotDay5Activity>

🔧 DESIGN ENGINEER, PRIMA AUTOMATION, IN

Jun 14 - Jan 16

- Programmed Siemens PLCs using TIA Portal and developed HMI screens for industrial automation projects.

- Designed electrical and control panels using AutoCAD.

- Created detailed electrical schematics and wiring diagrams using E-plan.